

Normoxic limitation of maximal oxygen consumption rate, aerobic scope and cardiac performance in exhaustively exercised rainbow trout (*Oncorhynchus mykiss*)

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Summary statement

Hyperoxia expands maximal oxygen consumption rate and aerobic scope in exhaustively exercise rainbow trout by increasing cardiac contractility and cardiac output

Abstract

In fish, maximum O₂ consumption rate (MO_{2max}) and aerobic scope can be expanded following exhaustive exercise in hyperoxia; however, the mechanisms explaining this are yet to be identified. Here, in exhaustively exercised rainbow trout (*Oncorhynchus mykiss*), we assessed the influence of hyperoxia on MO_{2max}, aerobic scope, cardiac function and blood parameters to address this knowledge gap. Relative to normoxia, MO_{2max} was 33% higher under hyperoxia, and this drove a similar increase in aerobic scope. Cardiac output, due to

increased stroke volume, was significantly elevated under hyperoxia at MO_{2max} indicating hyperoxia released a constraint on cardiac contractility apparent with normoxia. Thus, hyperoxia improved maximal cardiac performance, thereby enhancing tissue O_2 delivery and allowing a higher MO_{2max} . Venous blood O_2 partial pressure (P_vO_2) was elevated in hyperoxia at MO_{2max} , suggesting a contribution of improved luminal O_2 supply in enhanced cardiac contractility. Additionally, despite reduced haemoglobin and higher P_vO_2 , hyperoxia treated fish retained a higher arterio-venous O_2 content difference at MO_{2max} . This may have been possible due to hyperoxia offsetting declines in arterial oxygenation known to occur following exhaustive exercise in normoxia. If this occurs, increased contractility at MO_{2max} with hyperoxia may also relate to an improved O_2 supply to the compact myocardium via the coronary artery. Our findings show MO_{2max} and aerobic scope may be limited in normoxia following exhaustive exercise due to constrained maximal cardiac performance and highlight the need to further examine whether or not exhaustive exercise protocols are suitable for eliciting MO_{2max} and estimating aerobic scope in rainbow trout.

1. Introduction

In fish, exhaustive exercise (also referred to as strenuous exercise, burst swimming or chase stress) is a form of swimming powered mostly by anaerobically derived adenosine triphosphate (ATP) production within the highly differentiated white musculature (Kieffer, 2000). Before it becomes “exhaustive”, the powerful, swift bursts of movement produced by this form of swimming are undoubtedly ecologically and evolutionarily important, for example, in predator prey interactions (Harper and Blake, 1990; Webb, 1976) or in overcoming strong river flows in upstream migrations (Burnett et al., 2014; Hinch and Bratty, 2000). However, if anaerobically powered swimming is performed for too long, fish become exhausted and display a suite of severe physiological disturbances including, but not limited to, depletion of fermentable fuels, blood and tissue acidosis, lactate accumulation in the blood and white muscle, and ionic and osmotic imbalances (see reviews by Kieffer, 2000; Milligan, 1996; Wood, 1991a). Collectively, these disturbances generate an O_2 debt – the so called excess post-exercise O_2 consumption (EPOC) – that must be repaid following exhaustive exercise as a consequence of homeostasis being restored (Brett 1964; Scarabello et al., 1991; Zhang et al., 2018).

EPOC is typically characterised by whole animal mass-specific O_2 consumption rate (MO_2) that follows a pattern of exponential decay: a peak in MO_2 is observed immediately or soon after exhaustive exercise ceases, before it falls rapidly and then gradually returns to a resting rate over several hours. For example, after an immediate peak following exhaustive exercise, MO_2 returns to a resting rate (indicating EPOC has been repaid) in 3.5–4.5 h in juvenile rainbow trout (*Oncorhynchus mykiss*) (Scarabello et al., 1991). Longer duration EPOC (12–15 h), however, may be apparent when defining its endpoint as a return to a basal MO_2 (in fish termed standard metabolic rate; SMR) rather than one-off measures of routine MO_2 (Zhang et al., 2018). The MO_2 peak observed during EPOC has attracted much interest as a proxy measure of maximum metabolic rate in fish (Norin and Clark, 2016). This is because chasing a fish to exhaustion is a convenient, time efficient method of eliciting maximal O_2 consumption rate (MO_{2max}) in fishes in general, and also because it is often the only method suitable for doing so in sluggish, benthic species that will not swim in a flume (Norin and Clark, 2016). These estimates of MO_{2max} are important because together with estimates of SMR they are used to derive a measure of physiological performance known as aerobic scope (*i.e.*, the difference between MO_{2max} and SMR) (Clark et al., 2013). Although aerobic scope has long been used as a measure of physiological performance (Fry, 1947), it has risen to prominence in the past 15 years as a means of assessing the vulnerability of fish to climate warming (Claireaux and Lefrancois, 2007; Clark et al., 2013; Grans et al., 2014; Lefevre, 2016; McArley et al., 2017; Norin et al., 2014; Pörtner and Farrell, 2008).

It is well known that various environmental factors modulate aerobic scope. Indeed, the consistency, constraint or expansion of aerobic scope under different conditions (*e.g.*, at different temperatures) is a major reason it has been promoted as a metric of performance (Claireaux and Lefrancois, 2007; Clark et al., 2013; Pörtner and Farrell, 2008). Water oxygenation is one such environmental factor modulating aerobic scope. While there is a consensus that as O_2 levels fall below normoxia (*i.e.*, become hypoxic) aerobic scope will eventually be constrained due to a limitation of MO_{2max} (Claireaux and Lefrancois, 2007; Claireaux and Chabot, 2016; Mandic and Regan, 2018), it is unclear whether MO_{2max} and aerobic scope are constrained by O_2 availability under normoxia, because few studies have examined the influence of hyperoxia (O_2 levels above normoxia) on these parameters (reviewed in McArley et al., 2020). Two recent studies, however, provide evidence MO_{2max} and aerobic scope can be constrained under normoxia in fish. In European perch (*Perca fluviatilis*), MO_{2max} was ~92% higher under hyperoxia (200% air saturation) than normoxia

(Brijs et al., 2015). Also, in two species of triplefin fishes, the twister (*Belapiscis medius*) and common triplefin (*Forsterygion lapillum*), MO_{2max} was ~25% higher under hyperoxia (200% air saturation) than normoxia (McArley et al., 2018). In both these studies, elevated MO_{2max} under hyperoxia drove a corresponding expansion of aerobic scope, because with hyperoxia SMR was unchanged in perch and only slightly elevated in both triplefin species. Additionally, in the perch study, maximal routine MO_2 approximately doubled relative to normoxia at temperatures approaching thermal limits during acute thermal ramping (Brijs et al., 2015), and in the triplefin study, MO_{2max} was ~50% higher under hyperoxia when fish were exhaustively exercised after an acute temperature increase from 21°C to 29°C (McArley et al., 2018). While these studies clearly indicate an expansion of MO_{2max} with hyperoxia, as yet, the mechanisms that drive this have not been identified. Furthermore, whether an expansion of MO_{2max} following exhaustive exercise under hyperoxia is a common response in fish is unclear. This possibility has only been considered in one other study, which found MO_{2max} and aerobic scope unchanged with hyperoxia in common sole (*Solea solea*) (Lefrançois and Claireaux, 2003).

In this study, we added rainbow trout to the short list of fishes in which the influence of hyperoxia on MO_{2max} and aerobic scope following exhaustive exercise has been examined. The focus was to determine whether any changes (presumably an increase) in MO_{2max} and aerobic scope following exhaustive exercise under hyperoxia could be linked to changes in cardiac function. Some evidence that cardiac performance is improved under hyperoxia at times of high O_2 demand already exists. In European perch, Ekström et al. (2016) found cardiac output was higher under hyperoxia (200% air saturation) relative to normoxia at temperatures approaching thermal limits during acute thermal ramping. Although these authors did not measure MO_2 , the elevated cardiac output seen in their study corresponds well with the increased routine MO_2 at high temperatures observed in perch under hyperoxia by Brijs et al. (2015). As such, it was hypothesised that if MO_{2max} was expanded following exhaustive exercise with hyperoxia in rainbow trout, it would correspond with elevated cardiac output. Ekström et al. (2016) also observed that the partial pressure of oxygen in venous blood (P_vO_2) was elevated in hyperoxia across acute test temperatures, suggesting the possibility that hyperoxia improves luminal O_2 supply to the myocardium. To see if this was also the case in rainbow trout following exhaustive exercise, we assessed P_vO_2 by sampling venous blood from an implanted cannula. In addition to MO_2 , cardiac and P_vO_2 assessment, the influence of hyperoxia on the magnitude and recovery dynamics of physiological

disturbances resulting from exhaustive exercise was examined. This included haematological variables (haemoglobin and haematocrit), venous blood pH, plasma lactate and plasma osmolality. We also determined EPOC to assess whether hyperoxia affected the magnitude or the rate of repayment of the O₂ debt associated with exhaustive exercise.

2. *Materials and methods*

2.1. *Animals, holding conditions and experimental set-up*

The rainbow trout used in this study were of mixed sex and obtained from a commercial hatchery (Vänneåns Fiskodling AB, Halland, Sweden). At the time of experimentation, there was no difference in body mass between fish in the normoxic (896.5 ± 47.5 g) and hyperoxic (883.5 ± 68.57 g) treatment groups ($t=0.1518$, $df=17$, $P=0.88$). Prior to experimentation, the fish were held in 600 L tanks for a period of at least four weeks for laboratory acclimation. These tanks were supplied with air saturated, recirculated freshwater at $\sim 10^{\circ}\text{C}$ and maintained under a 12:12 h photoperiod. Fish were fed pellets (7 mm, Protec Trout pellets, Skretting, Norway) twice a week until the start of the experiments, but food was always withheld for a period of three days prior to experimentation. All experimental procedures were covered by ethical permit 165-2015, approved by the regional ethical committee in Gothenburg.

All measurements outlined in the following sections were completed with the fish housed in a respirometer (see section 2.4 below) held in a 120 L experimental tank. Two of these identical tanks, which each held an individual respirometer, were used to perform the experiments. The experimental tanks received temperature controlled freshwater (mean: $9.9 \pm 0.02^{\circ}\text{C}$, range: $9.7\text{-}10.8^{\circ}\text{C}$) from the main recirculated supply feeding the fish holding tanks. The normoxia treatment condition was maintained by bubbling the water supply to respirometers with air, while the hyperoxia treatment condition was maintained by bubbling pure O₂ at a set rate from a manually adjusted gas bottle (see section 2.3 for details).

2.2. Surgery and instrumentation

Surgery began by anaesthetising fish in 10°C freshwater containing 150 mg L⁻¹ MS-222 (Tricaine methanesulfonate, Scan Aqua AS, Årnes, Norway) buffered with 300 mg L⁻¹ NaHCO₃. Once ventilatory strokes ceased, mass was taken before placing fish right side up on a surgery table lined with foam. Throughout surgery, the gills were perfused with a continuous flow of aerated freshwater (10°C) containing 75 mg L⁻¹ MS-222 and 150 mg L⁻¹ NaHCO₃. Two procedures, which were completed in approximately 90 min, were carried out during surgery. Firstly, the ventral aorta was accessed via a small incision made in the isthmus within the right opercular cavity and freed from surrounding tissue by blunt dissection. A 2.5-mm Transonic transit-time blood flow probe (L type; Transonic Systems, Ithaca, NY) was then placed around the ventral aorta to allow recordings of ventral aortic blood flow (cardiac output). The flow probe was secured in place with two 3-0 silk sutures placed around the probe lead within the opercular cavity and a suture placed around the probe lead where it exited the opercular cavity. Additional silk sutures (2-0) positioned below the pectoral fin, below the lateral line ~5 cm posterior to the operculum, and adjacent to the dorsal fin fixed the probe lead to the fish's body. In the second part of surgery, the ductus of Cuvier was cannulated with a PE50 catheter to allow venous blood sampling as previously described by Sandblom et al. (2006). The catheter was filled with heparinised saline (25 IU/ml) and attached to the fish's body with silk sutures (3-0) positioned adjacent to where the catheter exited the opercula cavity and adjacent to the dorsal fin.

2.3. Experimental protocol

Following surgery, fish were transferred to the respirometers and immediately placed under treatment conditions of either normoxia or hyperoxia. The mean water O₂ level inside the respirometers throughout the duration of the experimental protocol was 95.9 ± 0.3 % air saturation (~20.1 kPa) and 210.2 ± 2.3 % air saturation (~44 kPa) in the normoxia and hyperoxia treatments, respectively. In the hyperoxic treatment, the maximum value inside the respirometers ranged from 207 to 230% air saturation (~43-48 kPa) among the ten replicates. The treatment groups are referred to as ~100% air saturation and ~200% air saturation throughout the remainder of this paper. After entering the respirometer, fish were left undisturbed for a post-surgery recovery period of ~22 h. MO₂ and cardiac function (cardiac

output and heart rate) were recorded continuously and a venous blood sample was drawn at the end of the post-surgery recovery period for assessment of pre-exhaustive exercise (from here referred to as pre-chase) baseline blood variables (see section 2.6 below). The fish was then disconnected from the recording equipment, removed from the respirometer and placed into a circular 50 L tank where it was exhaustively exercised by manual chasing (*i.e.*, repeated tail grabbing) for a period of 5 min. The transfer process to the chasing tank was completed in approximately 15 s. During the exhaustive exercise protocol, the water O₂ level in the tank was maintained at the level assigned to the respective treatment groups (*i.e.*, the normoxia group was chased in ~100% air saturation, and the hyperoxia group was chased in ~200% air saturation) and a temperature of ~10°C. All fish reached an exhausted state (*i.e.*, they no longer performed burst movements when the tail was grabbed) after 5 min chasing, and there were no obvious differences between treatment groups as to the speed at which fish became exhausted during chasing. After chasing, the fish was transferred back into the respirometer, reconnected to the recording equipment and post-exhaustive exercise (from here referred to as post-chase) respirometry cycles were initiated. This transfer process was always completed within 45 s of the completion of chasing. At this point, a venous blood sample was also drawn to assess blood variables immediately post-chase (0 h post-chase). MO₂ and cardiac function were then again measured continuously for a period of ~20 h, and further venous blood samples were taken at 1, 2, 4 and 20 h post-chase.

2.4. Respirometry and data acquisition for cardiac variables

MO₂ was measured using intermittent stop-flow respirometry (Steffensen, 1989). Respirometers (10 L volume) were constructed of a section of cylindrical PVC tube that was sealed with end caps fitted with O-rings. A spout placed in the middle of the respirometer, which extended above the water level of the experimental set-up, provided an exit for the catheter and flow probe lead. To flush the respirometer after each measurement cycle (see section 2.5), a submersible flush-pump (Eheim Universal 1200, Eheim, Deizisau, Germany) was connected to a fitting in the end cap via silicon tubing. The flush-pump was connected to a control switch coupled to a PowerLab system (ADInstruments, Castle Hill, Australia), which allowed the flush pump to be switched on and off automatically using LabChart pro data acquisition software (version 7.3.2, ADInstruments, Castle Hill, Australia). To continually mix water within the respirometer, an inline pump (Eheim Universal 1200) was

connected to a fitting in each end cap via a silicon tubing recirculation loop. A fibre optic O₂ probe (OXROB10, PyroScience, Aachen, Germany), sealed in a fitting placed in the recirculation loop, continually measured the water O₂ level (% air saturation) within the respirometer, and O₂ values were recorded using LabChart Pro data acquisition software via a Firesting O₂ meter (PyroScience, Aachen, Germany) connected to a PowerLab. To assess cardiac function, the flow probe (individually bench calibrated at 10°C according to the manufacturer's instructions) placed around the ventral aorta was connected to a Transonic blood flow meter (model T206; Transonic Systems, Ithaca, NY), and the signals were recorded at a sampling rate of 10 Hz using a PowerLab system and LabChart pro data acquisition software.

2.5. Calculation of cardiorespiratory variables

The slope of the linear decline in O₂ within the respirometer during measurement cycles (*i.e.*, when only the mixing pump was on) was used to determine MO₂ using the following formula:

$$MO_2 = [(V_r - V_f) \times (\Delta\%Sat/t) \times \alpha] / (M_b)$$

where V_r is the volume of the respirometer, V_f is the volume of the fish assuming that 1 g of fish equals 1 ml of water, Δ%Sat/t is the change in O₂ (% air saturation) per unit time, α is the temperature specific solubility coefficient of O₂ in freshwater and M_b is the body mass of the fish (Clark et al., 2013). The first 30-60 s of each measurement cycle was excluded to ensure only the linear section of the decline in O₂ was included in slope determinations (R² values for these slopes always remained above 0.98). Measurement cycles were interspersed with flush cycles to replace respirometer water. The first five cycles following exhaustive exercise always consisted of 90 s measurement periods and 3 minute flush periods; however, as the MO₂ of the fish recovered, the length of measurement and flush cycles were increased to allow a sufficient decline in O₂ during each cycle. This occurred in stages for approximately 4 h following exhaustive exercise, and then overnight measurement (~8-10 min) and flush (~5-7 min) cycles were initiated. The O₂ level in the respirometer remained above 85% air saturation in the normoxic treatment for >98% of the protocol and above 185% air saturation in the hyperoxic treatment for >99% of the protocol. In background O₂ consumption checks, a positive slope was detected, which was likely related to a small temperature increase

($\sim 0.10^{\circ}\text{C}$) inside the respirometer during measurement cycles. To account for this in estimates of fish MO_2 , the positive background slope determined at the start of the protocol was added to the slope value of each measurement cycle.

SMR was estimated as the mean of the lowest ten MO_2 values recorded during the ~ 20 h post-chase recovery period (Brijs et al., 2015; Norin et al., 2014). MO_2 values more than two standard deviations (*i.e.*, the standard deviation of the lowest ten MO_2 values for an individual fish) below a fish's SMR were removed. In total, nine MO_2 values were removed, and these were on average $0.25 \text{ mg O}_2 \text{ kg}^{-1} \text{ h}^{-1}$ (range: $0.05\text{--}0.65 \text{ mg O}_2 \text{ kg}^{-1} \text{ h}^{-1}$) below two standard deviations of SMR. $\text{MO}_{2\text{max}}$ was defined as the highest MO_2 value measured at any point post-chase and was seen within the first five post-chase measurement cycles in all fish. Aerobic scope was calculated as the difference between a fish's SMR and $\text{MO}_{2\text{max}}$. EPOC was calculated as the area under the curve bounded by post-chase MO_2 and SMR as a baseline. This was done using the area under the curve function in Graph Pad Prism (Version 8.4.2). The baseline value indicating when EPOC was completed was set as the highest MO_2 value used in the estimation of SMR for an individual fish (*i.e.*, the tenth lowest MO_2 value recorded post-chase). The MO_2 curves were smoothed to prevent periods of elevated MO_2 that likely resulted from increased activity from being included in EPOC calculation. Smoothing involved removing MO_2 values that increased by more than 5% relative to the previous MO_2 value until the baseline MO_2 threshold was reached (Zhang et al., 2018). In these instances, MO_2 values were removed until MO_2 returned to within 5% of the MO_2 value recorded immediately prior to the first removed value. The EPOC duration was defined as the time after exhaustive exercise at which MO_2 became equal to the tenth lowest MO_2 value used to estimate an individual fish's SMR. The rate of EPOC repayment was calculated as total EPOC divided by the EPOC duration.

Cardiac output was determined from blood flow data and normalised to body mass ($\text{mL min}^{-1} \text{ kg}^{-1}$), and heart rate was determined from the pulsatile blood flow measurements. Cardiac stroke volume ($\text{mL heart beat}^{-1}$) was calculated by dividing cardiac output by heart rate. The main purpose of the post-chase cardiac measurements was to identify whether improved heart performance was a driver of higher $\text{MO}_{2\text{max}}$ under hyperoxia. As such, the maximum values for cardiac output, heart rate and stroke volume presented here are taken from the same time when $\text{MO}_{2\text{max}}$ was measured. Resting values for cardiac output, heart rate and stroke volume were defined as the mean value for each variable over the ten lowest MO_2 measurement cycles (*i.e.*, they were tied to SMR). Scope for each of these cardiac variables was calculated

as the difference between the variable at MO_{2max} and the variable at SMR. Subsequently, coupled measurements of MO_2 and cardiac output were used to approximate the arterio-venous oxygen content difference (A-V O_2 content difference) according to rearrangement of the Fick equation:

$$\text{A-V } O_2 \text{ content difference} = MO_2 / \text{cardiac output}$$

Maximum, resting and scope for A-V O_2 content difference were defined as for cardiac variables (*i.e.*, tied to SMR and MO_{2max}).

To assess changes in MO_2 , cardiac variables and A-V O_2 content difference, pre-chase and post-chase routine values for each of these variables were determined for multiple time points throughout the protocol. The sampling time points were pre-chase (treated as a resting control value as described above), immediately post-chase (0 h) and 1, 2, 4, 8, 14 and 20 h post-chase. Routine values for each cardiorespiratory variable were defined as the mean value across two MO_2 measurement cycles at each of the predefined sampling time points. The exception to this was the 0 h cardiorespiratory measurements, which were always taken from the first MO_2 measurement cycle following exhaustive exercise.

2.6. Blood analysis

At most sampling time points (pre-chase and 0, 1, 2, 4 and 20 h post-chase), ~250 μ l of blood was drawn into a heparinised 1 mL syringe via the venous cannula. P_vO_2 was measured using a fibre optic O_2 probe (PyroScience, Aachen, Germany) calibrated at 10°C. To minimise exposure of the blood sample to air, the O_2 probe was pushed through the syringes rubber stopper and sealed in place with silicone. Thus, when a sample was taken, the blood came into contact with the O_2 probe tip sealed within the syringe. Once the blood sample was drawn, the catheter was removed from the syringe and the tip of the syringe was sealed. The syringe was then placed in 10°C water until the P_vO_2 signal plateaued, which typically occurred within ~3 min. The P_vO_2 signal was recorded via a Firesting O_2 meter connected to a PowerLab. After P_vO_2 was taken, the blood sample was transferred to a 1 mL centrifuge tube and placed in a 10°C water bath. Extracellular blood pH (pHe) was then measured using a handheld pH meter (Sentron SI400, Sentron Europe, Leek, Netherlands) calibrated at 10°C. Next, haematocrit (Hct) and haemoglobin (Hb) were measured in duplicate sub-samples. Hct was determined as the fractional red cell volume after centrifugation of blood in 80 μ l

microcapillary tubes at 10,000 rpm for 5 min, and Hb concentration was determined using a handheld Hb 201+ meter (Hemocue AB, Ängelholm, Sweden). The Hb values were corrected for fish blood following Clark et al. (2008). Mean corpuscular haemoglobin concentration (MCHC) was subsequently calculated as haemoglobin concentration/haematocrit $\times$ 100. The remaining blood was centrifuged at 10,000 rpm for 5 min, and the plasma was immediately frozen and stored at -80°C. The plasma samples were subsequently used to determine plasma lactate using a commercially available lactate assay kit (Lactate Colorimetric Assay Kit II, Biovision, California, USA). Osmolality was determined on thawed plasma sample using an osmometer (Model 3320 Osmometer, Advanced Instruments INC, Massachusetts, USA).

2.7. Statistics

All statistical analyses were performed using the IBM SPSS Statistics 26 software package, and significance was set at $P < 0.05$. SMR, MO_{2max} , and EPOC were compared between normoxia and hyperoxia using independent samples t-tests, and aerobic scope was compared between normoxia and hyperoxia using Welch's t-test. Cardiac output, heart rate, stroke volume and A-V O_2 content difference at MO_{2max} and SMR, as well as the scope for each of these variables, were compared between normoxia and hyperoxia with independent sample t-tests. For stroke volume scope, means were compared using Welch's t-test.

Mixed two-way analysis of variance (ANOVA) was used to compare routine MO_2 , routine heart function, routine A-V O_2 content difference and blood parameters between normoxia and hyperoxia pre and post-chase. Water O_2 level (normoxia or hyperoxia) was set as the between-subjects variable, and sampling time point was set as the within-subjects variable. Sampling time points included in the analysis of routine MO_2 , routine heart function and routine A-V O_2 content difference were pre-chase and 0, 1, 2, 4, 8, 14 and 20 h post-chase. Blood parameters were analysed using samples taken pre-chase and 0, 1, 2, 4 and 20 h post-chase. For the two-way mixed ANOVA of MO_2 , there was a violation of equality of variances among treatments at the 1 h (Levene's Test: $P = 0.012$) post-chase time point. Equality of variances could not be satisfied through data transformations, so the analysis was run with and without the 1 h time point included to determine if the outcome of the analysis was impacted. There was no difference in the conclusion drawn from the analysis when the 1 h time point was excluded, so the analysis including all time points is reported in the Results section. The same approach was taken for the two-way mixed ANOVA of stroke volume and

plasma lactate, where the exclusion of one time point (14 h for stroke volume and 2 h for plasma lactate) with unequal variances (Levene's Test: $P=0.017$ and $P=0.018$, respectively) among treatment groups had no influence on the conclusion drawn from the analysis. For the two-way mixed ANOVA of A-V O_2 content difference, the analysis was performed with the 20 h time point excluded to satisfy the assumption of equality of covariance matrices as assessed by a Box's M Test. P_vO_2 was analysed using natural log transformed data. Due to violations of equality of variance and equality of covariance matrices, which could not be corrected with transformations, the two-way mixed ANOVA for plasma osmolality was performed with the 1 h, 2 h and 20 h post-chase time points excluded. A second two-way mixed ANOVA was performed with the pre-chase and 20 h post-chase time points to confirm if recovery of osmolality was completed by the end of the protocol. In cases where there was a violation of sphericity (Mauchly's Test of Sphericity $P<0.05$), Greenhouse-Geisser adjusted F-Tests and P-values were interpreted in the analysis. Where an interaction was significant, simple main effects were assessed at each level of the within-subjects variable to determine if there were significant differences between normoxia and hyperoxia at a particular sampling time point. Additionally, in the case of a significant interaction, a repeated measures ANOVA with Bonferroni adjusted *post-hoc* comparisons was performed within each level of the between-subjects variable to determine if there were differences in post-chase recovery dynamics between normoxia and hyperoxia. As the main focus was determine if there were any differences between normoxia and hyperoxia in the rate of recovery to a resting condition post-chase, only the *post-hoc* comparisons between pre-chase values (treated as a resting control value) and post-chase values are reported in the Results. In the case of significant main effects without an interaction, Bonferroni adjusted *post-hoc* comparisons among sampling time points were made across the normoxia and hyperoxia treatments. Again, only the *post-hoc* comparisons between the pre-chase and post-chase values are reported in the Results. No *post-hoc* comparisons were required in the case where there was a significant main effect of water O_2 level because there were only two levels of this variable (*i.e.*, normoxia versus hyperoxia).

3. Results

3.1. Maximum cardiorespiratory responses and physiological status immediately following exhaustive exercise in normoxia and hyperoxia

$\text{MO}_{2\text{max}}$, which was observed within the first five post-chase respirometry cycles in all fish (*i.e.*, no more than 14 min post-chase), was 33% higher under hyperoxia than normoxia (hyperoxia: $391.6 \pm 24.8 \text{ mg O}_2 \text{ kg}^{-1} \text{ h}^{-1}$; normoxia: $294.1 \pm 12.2 \text{ mg O}_2 \text{ kg}^{-1} \text{ h}^{-1}$; $t=-3.40$, $\text{df}=17$, $P=0.003$; Fig. 1-F). A significantly higher MO_2 ($P<0.05$) was also seen with hyperoxia when only the 0 h post-chase time point (*i.e.*, the first post-chase MO_2 measurement) was considered (Fig. 2-A). Since SMR was similar between treatments (normoxia: $52.4 \pm 2.4 \text{ mg O}_2 \text{ kg}^{-1} \text{ h}^{-1}$; hyperoxia: $49.3 \pm 1.2 \text{ mg O}_2 \text{ kg}^{-1} \text{ h}^{-1}$; $t=1.01$, $\text{df}=17$, $P=0.33$, Fig. 1-A), the higher $\text{MO}_{2\text{max}}$ seen under hyperoxia translated to a significant elevation of aerobic scope under hyperoxia relative to normoxia (hyperoxia: $342.3 \pm 25.1 \text{ mg O}_2 \text{ kg}^{-1} \text{ h}^{-1}$; normoxia: $241.8 \pm 12.3 \text{ mg O}_2 \text{ kg}^{-1} \text{ h}^{-1}$; $t=-3.59$, $\text{df}=13.10$, $P=0.003$; Fig. 1-K). The higher $\text{MO}_{2\text{max}}$ under hyperoxia was mainly explained by improved cardiac function, because at $\text{MO}_{2\text{max}}$ cardiac output was also significantly elevated under hyperoxia (hyperoxia: $32.6 \pm 1.6 \text{ mL min}^{-1} \text{ kg}^{-1}$; normoxia: $27.1 \pm 2 \text{ mL min}^{-1} \text{ kg}^{-1}$; $t=-2.17$, $\text{df}=17$, $P=0.045$, Fig. 1-G). Higher cardiac output under hyperoxia at $\text{MO}_{2\text{max}}$ was mainly due to a trend for increased stroke volume (hyperoxia: $0.52 \pm 0.02 \text{ mL kg}^{-1}$; normoxia: $0.45 \pm 0.03 \text{ mL kg}^{-1}$; $t=-1.84$, $\text{df}=17$, $P=0.084$, Fig. 1-I), as heart rate was nearly identical at $\text{MO}_{2\text{max}}$ between treatments (normoxia: $60.7 \pm 1.5 \text{ beats min}^{-1}$; hyperoxia: $62.4 \pm 1.1 \text{ beats min}^{-1}$; $t=-0.95$, $\text{df}=17$, $P=0.36$, Fig. 1-H). An identical pattern was observed for these cardiac variables when only the 0 h post-chase time point was considered (Fig. 2-B, C&D). There was no difference in A-V O_2 content difference at $\text{MO}_{2\text{max}}$ between hyperoxia and normoxia (hyperoxia: $0.20 \pm 0.01 \text{ mg O}_2 \text{ mL}^{-1}$; normoxia: $0.19 \pm 0.01 \text{ mg O}_2 \text{ mL}^{-1}$; $t=-0.55$, $\text{df}=17$, $P=0.59$; Fig. 1-J), and this was also the case at the 0 h post-chase time point (Fig. 2-E). Moreover, at 0 h post-chase, close to the point when $\text{MO}_{2\text{max}}$ was observed, Hb and Hct were significantly lower ($P>0.05$) under hyperoxia than normoxia, but there was no difference in MCHC between treatments (Fig. 3). Additionally, $P_v\text{O}_2$ was significantly ($P<0.05$) elevated under hyperoxia relative to normoxia at 0 h post-chase (Fig. 4-A). No differences between treatments were observed for venous pHe, plasma lactate and plasma osmolality at the 0 h post chase time point (Fig. 4-B, C&D).

3.2. Cardiorespiratory function and physiological status at rest in normoxia and hyperoxia

At SMR, which was defined as the mean of the ten lowest MO_2 values measured post-chase, no differences existed between treatments in cardiac output (normoxia: $14 \pm 1.2 \text{ mL min}^{-1} \text{ kg}^{-1}$; hyperoxia: $14.1 \pm 1.1 \text{ mL min}^{-1} \text{ kg}^{-1}$; $t=-0.03$, $df=17$, $P=0.98$; Fig. 1-B), heart rate (normoxia: $50.9 \pm 2.8 \text{ beats min}^{-1}$; hyperoxia: $47.5 \pm 2.6 \text{ beats min}^{-1}$; $t=-0.03$, $df=17$, $P=0.98$; Fig. 1-C), stroke volume (normoxia: $0.28 \pm 0.03 \text{ mL kg}^{-1}$; hyperoxia: $0.30 \pm 0.02 \text{ mL kg}^{-1}$; $t=-0.46$, $df=17$, $P=0.65$; Fig. 1-D) and A-V O_2 content difference (normoxia: $0.066 \pm 0.006 \text{ mg O}_2 \text{ mL}^{-1}$; hyperoxia: $0.061 \pm 0.005 \text{ mg O}_2 \text{ mL}^{-1}$; $t=0.62$, $df=17$, $P=0.54$; Fig. 1-E). Similarly, there were no differences in routine MO_2 , cardiac variables and A-V O_2 content difference between treatments when fish were in a rested state pre-chase and 20 h post-chase (Fig. 2). Pre-chase and 20 h post-chase Hb and Hct, however, were significantly ($P<0.05$) depressed under hyperoxia relative to normoxia, while MCHC was similar between treatments at both times (Fig. 3). Alongside depressed Hb and Hct, $P_v\text{O}_2$ was significantly ($P<0.05$) elevated under hyperoxia relative to normoxia at both the pre-chase and 20 h post-chase time points (Fig. 4-A). Across treatments, however, no differences ($P>0.05$) in venous pHe, plasma lactate and plasma osmolality existed at these times (Fig. 4-B, C&D). Thus, at rest, both pre-chase and 20 h post-chase, the only differences observed between treatments were depressed Hb and Hct, and elevated $P_v\text{O}_2$ with hyperoxia.

3.3. Excess post-exercise oxygen consumption and recovery of cardiorespiratory function following exhaustive exercise in normoxia and hyperoxia

Despite higher $\text{MO}_{2\text{max}}$ with hyperoxia, there were no differences in total EPOC (normoxia: $530.4 \pm 26 \text{ mg O}_2 \text{ kg}^{-1}$; hyperoxia: $533.9 \pm 37.1 \text{ mg O}_2 \text{ kg}^{-1}$, $t=-0.07$, $df=17$, $P=0.94$; Fig. 5-A), EPOC duration (normoxia: $13.4 \pm 0.9 \text{ h}$; hyperoxia: $12.4 \pm 1.2 \text{ h}$, $t=0.72$, $df=17$, $P=0.48$; Fig. 5-B) and the rate of EPOC repayment (normoxia: $40.7 \pm 2.9 \text{ mg O}_2 \text{ kg}^{-1} \text{ h}^{-1}$; hyperoxia: $46.6 \pm 5.1 \text{ mg O}_2 \text{ kg}^{-1} \text{ h}^{-1}$, $t=-0.97$, $df=17$, $P=0.33$; Fig. 5-C) between treatments. When considering routine MO_2 , however, fish in hyperoxia recovered to a pre-chase level faster than did fish in normoxia; this occurred by 8 h post-chase in hyperoxia and by 14 h post-chase in normoxia (Fig. 2-A). Matching the pattern for routine MO_2 , post-chase routine cardiac output recovered to a pre-chase level faster under hyperoxia (by 4 h post-chase) than under normoxia (by 8 h post-chase) (Fig. 2-B). There were no differences between treatments

in the post-chase recovery dynamics of heart rate, stroke volume and A-V O₂ content difference (Fig. 2-C, D&E). Exhaustive exercise impacted Hb, Hct, MCHC, P_vO₂, venous pHe, plasma lactate and plasma osmolality similarly in hyperoxia and normoxia, such that there were no differences in recovery dynamics between treatments for any of these variables (Fig. 3 and Fig. 4).

4. Discussion

4.1. Hyperoxia increases maximum cardiorespiratory performance in rainbow trout following exhaustive exercise but does not influence repayment of O₂ debt

Here, in rainbow trout, MO_{2max} was 33% higher under hyperoxia than normoxia, and since SMR was essentially the same between treatments, this also drove a corresponding 41% increase of aerobic scope. At the same hyperoxic water O₂ level (~200% air saturation), the magnitude of increase in MO_{2max} and aerobic scope seen here was comparable to that seen in two triplefin fishes, ~25% and ~30% for MO_{2max} and aerobic scope, respectively (McArley et al., 2018), whereas it was less than the ~92% increase in MO_{2max} and aerobic scope seen in European perch (Brijs et al., 2015). Although MO_{2max} increased with hyperoxia, the total O₂ debt (EPOC) generated by exhaustive exercise – a highly anaerobic form of swimming (Milligan and Girard, 1993; Milligan, 1996) – was identical between normoxia and hyperoxia. The similarity of EPOC between oxygenation treatments, near identical profiles of lactate accumulation in the plasma and similar changes in venous blood pH, suggests the anaerobic requirements of exhaustive exercise were unmodified by hyperoxia. The energetic requirements of correcting osmotic imbalances caused by fluid and ion shifts between intracellular and extracellular compartments also make up a significant proportion of EPOC in fish (Wood, 1991a), but as indicated by near identical changes in plasma osmolality following exhaustive exercise, these, too, appeared unaffected by hyperoxia. Overall, it appears hyperoxia, despite increasing MO_{2max}, does not influence either the magnitude of O₂ debt generated from exhaustive exercise or the rate at which this O₂ debt is repaid in any meaningful way.

The elevated MO_{2max} and aerobic scope seen with hyperoxia in the current study was mainly the result of a 20% higher cardiac output. This was due to higher cardiac stroke volume, as heart rate was essentially identical at MO_{2max} between treatments. Thus, immediately after exhaustive exercise, hyperoxia released a constraint on cardiac contractility apparent with

normoxia, and this allowed, at least in part, for MO_2 to proceed at a higher rate. While it was confirmed that improved cardiac contractility following exhaustive exercise with hyperoxia was associated with elevated P_vO_2 , the magnitude of difference in P_vO_2 was relatively small (+0.8 kPa with hyperoxia), and it is questionable whether this would have driven the observed differences in contractility. In European perch subjected to thermal ramping, hyperoxia (200% air saturation) treated fish maintained higher routine cardiac output at temperatures approaching upper critical thermal limits (29°C–31°C) than did perch in normoxia (Ekström et al., 2016). In the same study, P_vO_2 was elevated with hyperoxia, which the authors suggested may have led to improved cardiac performance through an enhanced luminal O_2 supply to the myocardium. It should be noted, however, that at the temperatures where differences in cardiac output emerged under hyperoxia in perch (29°C–30°C), a greater difference in P_vO_2 between hyperoxia and normoxia was observed than that seen in rainbow trout in the current study (in perch, +3.9 kPa and +1.7 kPa with hyperoxia at 29°C and 30°C, respectively, and in the current study, +0.8 kPa with hyperoxia following exhaustive exercise).

In salmonids, the outer compact layer of the heart (compact myocardium) is supplied directly with oxygenated blood from the gills via the coronary artery. This contrasts with approximately 2/3 of all teleost species, including European perch, in which the heart only receives O_2 via the venous blood returning to the heart (Ekström et al., 2016; Ekström et al., 2017; Farrell et al., 2012). Thus, here, in addition to the slightly elevated P_vO_2 with hyperoxia, it is also possible that hyperoxia enhanced the coronary arterial O_2 supply to the compact myocardium, which in turn may have contributed to improved contractility. All but one study assessing the influence of hyperoxia on blood oxygenation in fish, including several in rainbow trout, show arterial blood O_2 partial pressure (P_aO_2) increases during environmental hyperoxia exposure (McArley et al., 2020). It is likely, therefore, that P_aO_2 was increased with hyperoxia in the current study. At rest under normoxia, P_aO_2 is normally around 14.5–18.0 kPa in rainbow trout (e.g. Kiceniuk and Jones, 1977; Morgenroth et al., 2019; Wang et al., 1994; Wood and Jackson, 1980). If such levels of P_aO_2 remained after exhaustive exercise, then any increase in P_aO_2 with hyperoxia leading to improved cardiac output might suggest an *in vivo* limitation of maximal contractility, presumably within the compact myocardium, at normoxic P_aO_2 . It is possible, however, that hyperoxia does not increase P_aO_2 above a typical resting normoxic level following exhaustive exercise. Indeed, several studies in rainbow trout have demonstrated P_aO_2 can fall to around ~8 kPa

immediately following exhaustive exercise (Ferguson and Tufts, 1992; Milligan and Wood, 1987; Primmitt et al., 1986; Wang et al., 1994). Therefore, a window may exist after exhaustive exercise where hyperoxia would not simply elevate P_aO_2 beyond the typical normoxic range (~ 16 kPa), but instead prevent it from collapsing and becoming mildly hypoxic, as seems to be the case in normoxia. Further investigation, however, is required to evaluate whether there are differences in arterial blood oxygenation in fish undergoing exhaustive exercise under hyperoxia or normoxia.

4.2. Hyperoxia may safeguard arterial haemoglobin oxygen saturation following exhaustive exercise

The A-V O_2 content difference (estimated by the Fick equation) at MO_{2max} was 6.3% higher under hyperoxia than normoxia. Although this difference was not significant, combined with higher cardiac output, it may have contributed to the higher MO_{2max} in hyperoxia. The increased A-V O_2 content difference was apparent despite Hb being, on average, 12.5 g L^{-1} lower under hyperoxia immediately post-chase, which would tend to reduce arterial O_2 content, and P_vO_2 being elevated under hyperoxia immediately post-chase, which would tend to increase venous O_2 content (*i.e.*, through increasing HbO_2 saturation). Together, in a situation where Hb was fully saturated in arterial blood and assuming the HbO_2 dissociation curve was similarly shaped with hyperoxia and normoxia, this should have caused a reduced A-V O_2 content difference under hyperoxia relative to normoxia. There are two possibilities that could explain this seemingly impossible outcome. Firstly, following exhaustive exercise, it may be that HbO_2 saturation in arterial blood is higher under hyperoxia than normoxia. As noted above, it is known that P_aO_2 can fall to around ~ 8 kPa following exhaustive exercise in normoxia, and this, along with possible contributions of Bohr/Root effects associated with blood acidosis, can cause reductions in HbO_2 saturation in arterial blood (Ferguson and Tufts, 1992; Milligan and Wood, 1987). As hyperoxia increases P_aO_2 in rainbow trout (Wood and Jackson 1980), it is possible hyperoxia offsets declines in P_aO_2 following exhaustive exercise in normoxia, thereby promoting higher HbO_2 saturation and a higher arterial O_2 content. Duthie and Hughes (1987) demonstrated there is no difference in MO_{2max} between hyperoxia and normoxia treated rainbow trout swum at critical swimming speed (U_{crit}). While this may seem paradoxical in relation to the elevated MO_{2max} observed in hyperoxia following exhaustive exercise in the present study, one reason to explain the lack of an effect of

hyperoxia in Duthie and Hughes (1987) could be that a reduction in PaO_2 and HbO_2 saturation is less likely to occur at U_{crit} under normoxia when swimming fish are able to ram ventilate. Indeed, Kiceniuk and Jones (1977) showed that PaO_2 , HbO_2 saturation and arterial O_2 content all remained similar to resting levels in rainbow trout swimming at U_{crit} under normoxia. Contrastingly, also in rainbow trout, Neumann et al. (1983) showed a significant decline in arterial O_2 content relative to resting levels – presumably linked to lower PaO_2 and HbO_2 saturation – following exhaustive exercise. If a reduction in arterial HbO_2 saturation following exhaustive exercise is common in fish and does not occur when fish are swimming (and ram ventilating), this could offer an explanation as to why, at least in some cases, higher $\text{MO}_{2\max}$ values are seen in fish swum at U_{crit} than in fish following exhaustive exercise (Raby et al., 2020).

The second possible reason explaining a higher A-V O_2 content difference with hyperoxia relates to a limitation with using the Fick equation (*i.e.*, dividing MO_2 by cardiac output) to estimate this parameter. In doing so, it is assumed that the total MO_2 of the organism is only the product of the O_2 delivered to the tissues via arterial blood multiplied by the rate of arterial blood flow (*i.e.*, cardiac output). In fish, however, this assumption is erroneous because O_2 can be consumed directly from the water by the skin and gills (Farrell et al., 2014; Feder and Burggren, 1985; Zena et al., 2017). The proportion of total MO_2 consumed directly by the skin in rainbow trout, for example, is estimated to be 15% under normoxia (Kirsch and Nonnotte, 1977). While it has not been evaluated in rainbow trout, the proportion of total MO_2 consumed directly by the skin has been shown to increase under hyperoxia in other fishes, for example, in European eels – *Anguilla anguilla* (Kirsch and Nonnotte, 1977; Le Moigne et al., 1986) and common carp – *Cyprinus carpio* (Takeda, 1989). An important distinction to make, however, is that, although the proportion of O_2 consumed directly by the skin and gills increases with hyperoxia – presumably because of increased O_2 diffusion gradients between hyperoxic water and cutaneous tissues – total MO_2 remains unchanged (Kirsch and Nonnotte, 1977; Le Moigne et al., 1986; Takeda, 1989). Thus, in itself, hyperoxia does not appear to increase total O_2 demand. If the proportion of total MO_2 attributed to direct O_2 uptake by the skin and gills was higher under hyperoxia than normoxia in the current study, then the estimates of A-V O_2 content difference derived from rearrangement of the Fick equation would be overestimated for the hyperoxic treatment, thereby inflating estimates relative to normoxia. In reality, the estimate of A-V O_2 content difference calculated here is actually an estimate of the total amount of O_2 removed from the

water via any route per unit of blood volume pumped by the heart, and direct measures of arterial and venous O₂ content alongside cardiac output and MO₂ are required to pinpoint where the O₂ consumed by the organism is being taken up and in what proportions. In sockeye salmon (*Oncorhynchus nerka*), although not in relation to hyperoxia, an analysis of this type has been completed (Farrell et al., 2014). It revealed the proportion of total MO₂ attributed to direct uptake by cutaneous tissues ranges from 12-48% and does not necessarily decline when metabolic rate increases, as has been previously assumed (Farrell et al., 2014). These findings show the potential for substantial and variable direct O₂ uptake by cutaneous tissues in salmonids, which in turn highlights the need for caution when interpreting parameters, such as the A-V O₂ content difference, derived from the Fick equation.

4.3. Hyperoxia induces reduced haemoglobin and haematocrit

The Hb concentration and Hct of blood were reduced in hyperoxia relative to normoxia under resting conditions, and this was apparent both before and following recovery from exhaustive exercise. In both normoxia and hyperoxia, Hb and Hct increased after exhaustive exercise, but here, too, the haematological differences seen between treatments remained. MCHC was also similar among treatments at rest and following exhaustive exercise. The observed pattern of a reduction in MCHC after exhaustive exercise accompanied by increases in Hb and Hct indicate a combination of splenic release of stored red blood cells, which boosts blood O₂ carrying capacity (Brijs et al., 2020; Kita and Itazawa, 1989; Pearson and Stevens, 1991; Yamamoto et al., 1980), and red blood cell swelling, which serves to protect blood O₂ transport through regulation of intracellular red blood cell pH and dilution of allosteric modifiers of Hb (Milligan and Wood, 1987; Nikinmaa, 1986). Both these responses (*i.e.*, splenic release and red blood cell swelling) are linked to the action of neurally and hormonally released catecholamines (*i.e.*, adrenaline and noradrenaline) (Kita and Itazawa, 1989; Nikinmaa, 1986), the latter which increase substantially in the plasma of exhaustively exercised rainbow trout (Milligan and Wood, 1987; Wood et al., 1990). It is also likely that part of the observed hemoconcentration following exhaustive exercise resulted from a reduction in plasma volume associated with fluid shifts from extracellular to intracellular fluid compartments (Pearson and Stevens, 1991; Wood, 1991a). The sharp increase in plasma osmolality following exhaustive exercise indicates such a fluid shift occurred.

The general reduction of Hb and Hct with hyperoxia suggests that after only ~22 h exposure to elevated water O₂ levels, when the first blood sample was taken, red blood cells had been removed from the circulation. A dilution of the blood, for example, due to possible fluid shifts associated with respiratory acidosis and bicarbonate accumulation in plasma during hyperoxia (Wood, 1991b), could also have caused Hb and Hct to decrease, but this seems unlikely as there were no differences in resting plasma osmolality or overall cardiovascular status between treatments. Similarly, in Senegalese sole (*Solea senegalensis*), lower Hb levels accompanied by lower red blood cell numbers were observed after 24 h exposure to 200% air saturation (Machado et al., 2018). Other studies involving short duration (1-4 days) hyperoxia, however, have not reported similar haematological changes (Brauner et al., 2000; Ekström et al., 2016; Karlsson et al., 2011; Mustafa et al., 2011; Takeda, 1990; Vanlandeghem et al., 2010). More often, the haematological responses to long term (weeks to months) hyperoxia have been assessed in fish (reviewed in McArley et al., 2020), and several studies in rainbow trout, specifically, have demonstrated either reduced Hb or Hct, or both in such circumstances (Dabrowski et al., 2004; Edsall and Smith, 1990; Jewett et al., 1991; Ritola et al., 2002). Thus, although the reduction in Hb and Hct with hyperoxia observed in the current study is not unique in rainbow trout, it is, to the best of our knowledge, the first to show these haematological changes can occur so rapidly. Assessing the role of the spleen, the known storage organ for red blood cells in fish (Fänge and Nilsson, 1985), in mediating changes in Hb and Hct during hyperoxia exposure may provide a useful avenue to begin understanding exactly how these acute haematological responses occur.

4.4. Conclusion

Our findings show, at least in rainbow trout, that one way hyperoxia can increase MO_{2max} and aerobic scope following exhaustive exercise is to improve cardiac contractility, thereby allowing higher cardiac output. An improvement of maximal cardiac performance alone, however, did not fully explain the higher MO_{2max} seen with hyperoxia. Somehow, despite a significantly reduced Hb level and higher P_vO₂, fish in hyperoxia maintained a higher A-V O₂ content difference as estimated by the Fick equation. One way this might have been possible is that hyperoxia offset declines in arterial HbO₂ saturation following exhaustive exercise that are known to occur under normoxia. Although somewhat speculative, this could raise arterial blood O₂ content higher under hyperoxia relative to normoxia, thus driving a higher A-V O₂

content difference. Future studies assessing arterial P_aO_2 , arterial HbO_2 saturation and arterial O_2 content would further elucidate exactly how hyperoxia expands MO_{2max} following exhaustive exercise and could have important implications for the utility of using exhaustive exercise to elicit MO_{2max} and estimate aerobic scope in fish.

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Competing Interests

The authors declare no competing interests.

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Data availability

Data is publicly available on figshare under the title “MS_data Normoxic limitation of maximal oxygen consumption rate, aerobic scope and cardiac performance in exhaustively exercised rainbow trout (*Oncorhynchus mykiss*)” (<https://doi.org/10.6084/m9.figshare.14815818>).

References

- Brauner, C. J., Seidelin, M., Madsen, S. and Jensen, F. B.** (2000). Effects of freshwater hyperoxia and hypercapnia and their influences on subsequent seawater transfer in Atlantic salmon (*Salmo salar*) smolts. *Can. J. Fish. Aquat. Sci.* **57**, 2054-2064.
- Brett, J. R.** (1964). The respiratory metabolism and swimming performance of young sockeye salmon. *J. Fish. Res. Board. Can.* **21**, 1183-1226.
- Brijs, J., Axelsson, M., Rosengren, M., Jutfelt, F. and Grans, A.** (2020). Extreme blood-boosting capacity of an Antarctic fish represents an adaptation to life in a sub-zero environment. *J. Exp. Biol.* **223**, 10.1242/jeb.218164.
- Brijs, J., Jutfelt, F., Clark, T. D., Grans, A., Ekstrom, A. and Sandblom, E.** (2015). Experimental manipulations of tissue oxygen supply do not affect warming tolerance of European perch. *J. Exp. Biol.* **218**, 2448-2454.
- Burnett, N. J., Hinch, S. G., Braun, D. C., Casselman, M. T., Middleton, C. T., Wilson, S. M. and Cooke, S. J.** (2014). Burst swimming in areas of high flow: delayed consequences of anaerobiosis in wild adult sockeye salmon. *Physiol. Biochem. Zool.* **87**, 587-598.
- Claireaux, G. and Chabot, D.** (2016). Responses by fishes to environmental hypoxia: integration through Fry's concept of aerobic metabolic scope. *J. Fish Biol.* **88**, 232-251.
- Claireaux, G. and Lefrancois, C.** (2007). Linking environmental variability and fish performance: integration through the concept of scope for activity. *Philos. Trans. R. Soc. Lond. B. Biol. Sci.* **362**, 2031-2041.

Clark, T. D., Eliason, E., Sandblom, E., Hinch, S. and Farrell, A. (2008). Calibration of a hand-held haemoglobin analyser for use on fish blood. *J. Fish Biol.* **73**, 2587-2595.

Clark, T. D., Sandblom, E. and Jutfelt, F. (2013). Aerobic scope measurements of fishes in an era of climate change: respirometry, relevance and recommendations. *J. Exp. Biol.* **216**, 2771-2782.

Dabrowski, K., Lee, K., Guz, L., Verlhac, V. and Gabaudan, J. (2004). Effects of dietary ascorbic acid on oxygen stress (hypoxia or hyperoxia), growth and tissue vitamin concentrations in juvenile rainbow trout (*Oncorhynchus mykiss*). *Aquaculture* **233**, 383-392.

Duthie, G. G. and Hughes, G. M. (1987). The effects of reduced gill area and hyperoxia on the oxygen consumption and swimming speed of rainbow trout. *J. Exp. Biol.* **127**, 349-354.

Edsall, D. A. and Smith, C. E. (1990). Performance of rainbow trout and Snake River cutthroat trout reared in oxygen-supersaturated water. *Aquaculture* **90**, 251-259.

Ekström, A., Axelsson, M., Gräns, A., Brijs, J. and Sandblom, E. (2017). Influence of the coronary circulation on thermal tolerance and cardiac performance during warming in rainbow trout. *Am. J. Physiol. Regul. Integr. Comp. Physiol.* **312**, R549-R558.

Ekström, A., Brijs, J., Clark, T. D., Gräns, A., Jutfelt, F. and Sandblom, E. (2016). Cardiac oxygen limitation during an acute thermal challenge in the European perch: effects of chronic environmental warming and experimental hyperoxia. *Am. J. Physiol. Regul. Integr. Comp. Physiol.* **311**, R440-R449.

Fänge, R. and Nilsson, S. (1985). The fish spleen: structure and function. *Experientia* **41**, 152-158.

Farrell, A. P., Eliason, E. J., Clark, T. D. and Steinhausen, M. F. (2014). Oxygen removal from water versus arterial oxygen delivery: calibrating the Fick equation in Pacific salmon. *J. Comp. Physiol. B.* **184**, 855-864.

Farrell, A. P., Farrell, N. D., Jourdan, H. and Cox, G. K. (2012). A perspective on the evolution of the coronary circulation in fishes and the transition to terrestrial life. In *Ontogeny and Phylogeny of the Vertebrate Heart* (ed. D. Sedmera and T. Wang), pp. 75-102. New York: Springer-Verlag.

Farrell, A. P. (2016). Pragmatic perspective on aerobic scope: peaking, plummeting, pejus and apportioning. *J. Fish Biol.* **88**, 322-343.

Feder, M. E. and Burggren, W. W. (1985). Cutaneous gas exchange in vertebrates: design, patterns, control and implications. *Biol. Rev.* **60**, 1-45.

Ferguson, R. and Tufts, B. (1992). Physiological effects of brief air exposure in exhaustively exercised rainbow trout (*Oncorhynchus mykiss*): implications for "catch and release" fisheries. *Can. J. Fish. Aquat. Sci.* **49**, 1157-1162.

Fry, F. E. J. (1947). Effects of the environment on animal activity. *Publ. Ontario Fish. Res. Lab.* **68**, 1-52.

Grans, A., Jutfelt, F., Sandblom, E., Jonsson, E., Wiklander, K., Seth, H., Olsson, C., Dupont, S., Ortega-Martinez, O., Einarsdottir, I. et al. (2014). Aerobic scope fails to explain the detrimental effects on growth resulting from warming and elevated CO₂ in Atlantic halibut. *J. Exp. Biol.* **217**, 711-717.

Harper, D. G. and Blake, R. W. (1990). Fast-start performance of rainbow trout *Salmo gairdneri* and northern pike *Esox lucius*. *J. Exp. Biol.* **150**, 321-342.

Hinch, S. G. and Bratty, J. (2000). Effects of swim speed and activity pattern on success of adult sockeye salmon migration through an area of difficult passage. *Trans. Am. Fish. Soc.* **129**, 598-606.

Jewett, M. G., Behmer, D. J. and Johnson, G. H. (1991). Effects of hyperoxic rearing water on blood hemoglobin and hematocrit levels of rainbow trout. *J. Aquat. Anim. Health.* **3**, 153-160.

Karlsson, A., Rosseland, B., Thorarensen, H. and Kiessling, A. (2011). Changes in arterial oxygen tension and physiological status in resting, unrestrained Arctic charr *Salvelinus alpinus* (L.) exposed to mild hypoxia and hyperoxia. *J. Fish Biol.* **78**, 962-966.

Kiceniuk, J. W. and Jones, D. R. (1977). The oxygen transport system in trout (*Salmo gairdneri*) during sustained exercise. *J. Exp. Biol.* **69**, 247-260.

Kieffer, J. D. (2000). Limits to exhaustive exercise in fish. *Comp. Biochem. Physiol. Part A Mol. Integr. Physiol.* **126**, 161-179.

Kirsch, R. and Nonnotte, G. (1977). Cutaneous respiration in three freshwater teleosts. *Respir. Physiol.* **29**, 339-354.

Kita, J. and Itazawa, Y. (1989). Release of erythrocytes from the spleen during exercise and splenic constriction by adrenaline infusion in the rainbow trout. *Jap. J. Ichthyol.* **36**, 48-52.

Le Moigne, J., Soulier, P., Peyraud-Waitzenegger, M. and Peyraud, C. (1986). Cutaneous and gill O₂ uptake in the European eel (*Anguilla anguilla* L.) in relation to ambient PO₂, 10–400 Torr. *Respir. Physiol.* **66**, 341-354.

Lefevre, S. (2016). Are global warming and ocean acidification conspiring against marine ectotherms? A meta-analysis of the respiratory effects of elevated temperature, high CO₂ and their interaction. *Conserv. Physiol.* **4**, cow009.

Lefrançois, C. and Claireaux, G. (2003). Influence of ambient oxygenation and temperature on metabolic scope and scope for heart rate in the common sole *Solea solea*. *Mar. Ecol. Prog. Ser.* **259**, 273-284.

Machado, M., Malheiro, D., Couto, A., Wilson, J. M., Guerreiro, M., Azeredo, R., Svendsen, J. C., Afonso, A., Serradeiro, R. and Costas, B. (2018). Acute hyperoxia induces systemic responses with no major changes in peripheral tissues in the Senegalese sole (*Solea senegalensis* Kaup, 1858). *Fish Shellfish Immunol.* **74**, 260-267.

Mandic, M. and Regan, M. D. (2018). Can variation among hypoxic environments explain why different fish species use different hypoxic survival strategies? *J. Exp. Biol.* **221**.

McArley, T. J., Hickey, A. J. R. and Herbert, N. A. (2018). Hyperoxia increases maximum oxygen consumption and aerobic scope of intertidal fish facing acutely high temperatures. *J. Exp. Biol.* **221**.

McArley, T. J., Sandblom, E. and Herbert, N. A. (2021). Fish and hyperoxia—From cardiorespiratory and biochemical adjustments to aquaculture and ecophysiology implications. *Fish Fish.* **22**, 324–355.

McArley, T. J., Hickey, A. J. and Herbert, N. A. (2017). Chronic warm exposure impairs growth performance and reduces thermal safety margins in the common triplefin fish (*Forsterygion lapillum*). *J. Exp. Biol.* **220**, 3527-3535.

Milligan, C. L. (1996). Metabolic recovery from exhaustive exercise in rainbow trout. *Comp. Biochem. Physiol. Part A Mol. Integr. Physiol.* **113**, 51-60.

Milligan, C. L. and Girard, S. S. (1993). Lactate metabolism in rainbow trout. *J. Exp. Biol.* **180**, 175-193.

Milligan, C. L. and Wood, C. M. (1987). Regulation of blood oxygen transport and red cell pHi after exhaustive activity in rainbow trout (*Salmo gairdneri*) and starry flounder (*Platichthys stellatus*). *J. Exp. Biol.* **133**, 263-282.

Morgenroth, D., Ekström, A., Hjelmstedt, P., Gräns, A., Axelsson, M. and Sandblom, E. (2019). Hemodynamic responses to warming in euryhaline rainbow trout: implications of the osmo-respiratory compromise. *J. Exp. Biol.* **222**,.

Mustafa, S. A., Al-Subiai, S. N., Davies, S. J. and Jha, A. N. (2011). Hypoxia-induced oxidative DNA damage links with higher level biological effects including specific growth rate in common carp, *Cyprinus carpio* L. *Ecotoxicology* **20**, 1455-1466.

Neumann, P., Høle, G. and Heisler, N. (1983). Cardiac output and regional blood flow in gills and muscles after exhaustive exercise in rainbow trout (*Salmo gairdneri*). *J. Exp. Biol.* **105**, 1-14.

Nikinmaa, M. (1986). Control of red cell pH in teleost fishes. *Ann. Zool. Fennici.* **23**, 223-235.

Norin, T. and Clark, T. D. (2016). Measurement and relevance of maximum metabolic rate in fishes. *J. Fish Biol.* **88**, 122-151.

- Norin, T., Malte, H. and Clark, T. D.** (2014). Aerobic scope does not predict the performance of a tropical eurythermal fish at elevated temperatures. *J. Exp. Biol.* **217**, 244-251.
- Pearson, M. and Stevens, E.** (1991). Size and hematological impact of the splenic erythrocyte reservoir in rainbow trout, *Oncorhynchus mykiss*. *Fish Physiol. Biochem.* **9**, 39-50.
- Pörtner, H. O., Bock, C. and Mark, F. C.** (2017). Oxygen- and capacity-limited thermal tolerance: bridging ecology and physiology. *J. Exp. Biol.* **220**, 2685-2696.
- Pörtner, H. O. and Farrell, A. P.** (2008). Ecology. Physiology and climate change. *Science* **322**, 690-692.
- Primmitt, D. R., Randall, D. J., Mazeaud, M. and Boutilier, R. G.** (1986). The role of catecholamines in erythrocyte pH regulation and oxygen transport in rainbow trout (*Salmo gairdneri*) during exercise. *J. Exp. Biol.* **122**, 139-148.
- Raby, G. D., Doherty, C. L., Mokdad, A., Pitcher, T. E. and Fisk, A. T.** (2020). Post-exercise respirometry underestimates maximum metabolic rate in juvenile salmon. *Conserv. Physiol.* **8**, coaa063.
- Ritola, O., Tossavainen, K., Kiuru, T., Lindström- Seppä, P. and Mölsä, H.** (2002). Effects of continuous and episodic hyperoxia on stress and hepatic glutathione levels in one-summer-old rainbow trout (*Oncorhynchus mykiss*). *J. Appl. Ichthyol.* **18**, 159-164.

- Sandblom, E., Axelsson, M. and McKenzie, D. J.** (2006). Venous responses during exercise in rainbow trout, *Oncorhynchus mykiss*: α -adrenergic control and the antihypotensive function of the renin–angiotensin system. *Comp. Biochem. Physiol. Part A Mol. Integr. Physiol.* **144**, 401-409.
- Scarabello, M., Heigenhauser, G. and Wood, C.** (1991). The oxygen debt hypothesis in juvenile rainbow trout after exhaustive exercise. *Respir. Physiol.* **84**, 245-259.
- Steffensen, J. F.** (1989). Some errors in respirometry of aquatic breathers: how to avoid and correct for them. *Fish Physiol. Biochem.* **6**, 49-59.
- Takeda, T.** (1989). Cutaneous and gill O₂ uptake in the carp, *Cyprinus carpio*, as a function of ambient PO₂. *Comp. Biochem. Physiol. Part A Mol. Integr. Physiol.* **94**, 205-208.
- Takeda, T.** (1990). Ventilation, cardiac output and blood respiratory parameters in the carp, *Cyprinus carpio*, during hyperoxia. *Respir. Physiol.* **81**, 227-239.
- Vanlandeghem, M., Wahl, D. H. and Suski, C.** (2010). Physiological responses of largemouth bass to acute temperature and oxygen stressors. *Fish. Manage. Ecol.* **17**, 414-425.
- Wang, Y., Heigenhauser, G. J. and Wood, C. M.** (1994). Integrated responses to exhaustive exercise and recovery in rainbow trout white muscle: acid-base, phosphogen, carbohydrate, lipid, ammonia, fluid volume and electrolyte metabolism. *J. Exp. Biol.* **195**, 227-258.
- Webb, P. W.** (1976). The effect of size on the fast-start performance of rainbow trout *Salmo gairdneri*, and a consideration of piscivorous predator-prey interactions. *J. Exp. Biol.* **65**, 157-177.

Wood, C. M. (1991a). Acid-base and ion balance, metabolism, and their interactions, after exhaustive exercise in fish. *J. Exp. Biol.* **160**, 285-308.

Wood, C. M. (1991b). Branchial ion and acid-base transfer in freshwater teleost fish: environmental hyperoxia as a probe. *Physiol. Zool.* **64**, 68-102.

Wood, C. M. and Jackson, E. B. (1980). Blood acid-base regulation during environmental hyperoxia in the rainbow trout (*Salmo gairdneri*). *Respir. Physiol.* **42**, 351-372.

Wood, C. M., Walsh, P. J., Thomas, S. and Perry, S. F. (1990). Control of red blood cell metabolism in rainbow trout after exhaustive exercise. *J. Exp. Biol.* **154**, 491-507.

Yamamoto, K., Yasuo, I. and Hiroshi, K. (1980). Supply of erythrocytes into the circulating blood from the spleen of exercised fish. *Comp. Biochem. Physiol. Part A Mol. Integr. Physiol.* **65**, 5-11.

Zena, L. A., Bícago, K. C., da Silva, G. S., Giusti, H., Glass, M. L. and Sanchez, A. P. (2017). Acute effects of temperature and hypercarbia on cutaneous and branchial gas exchange in the South American lungfish, *Lepidosiren paradoxa*. *J. Therm. Biol.* **63**, 112-118.

Zhang, Y., Claireaux, G., Takle, H., Jørgensen, S. and Farrell, A. (2018). A three-phase excess post-exercise oxygen consumption in Atlantic salmon *Salmo salar* and its response to exercise training. *J. Fish Biol.* **92**, 1385-1403.

Figures

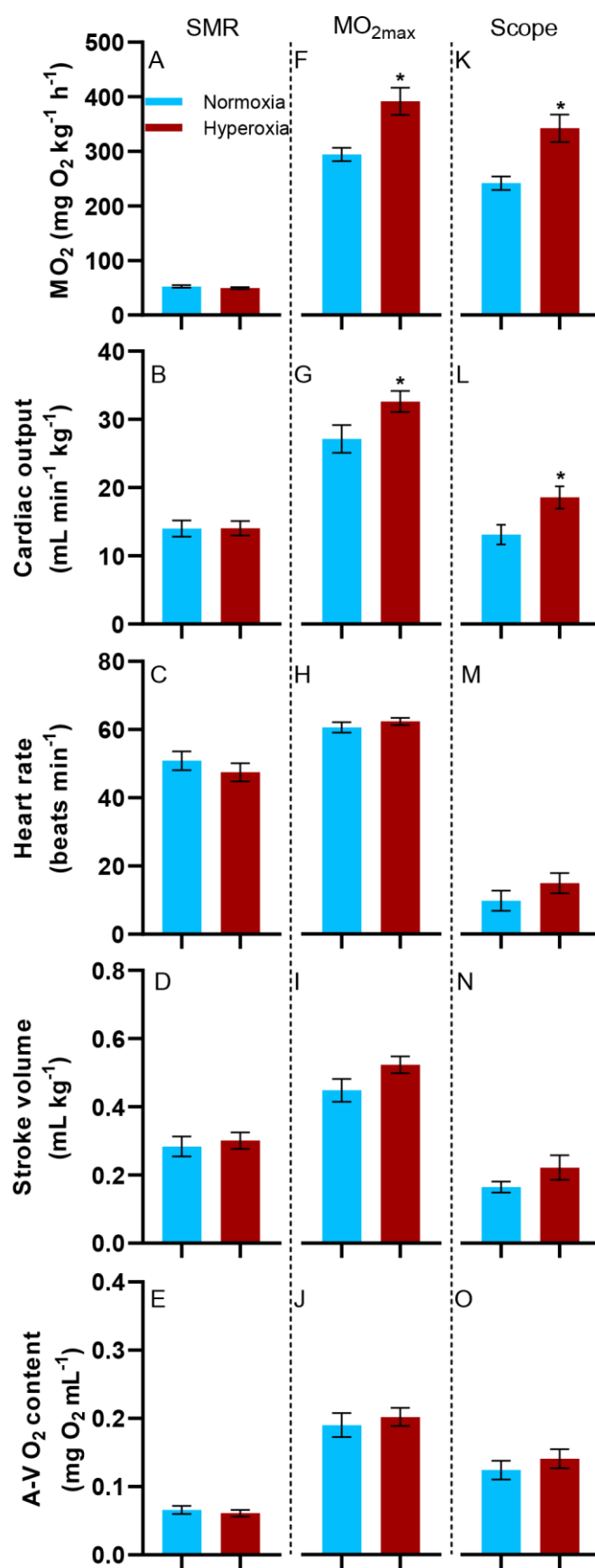


Fig. 1 Cardiorespiratory function in rainbow trout (*Oncorhynchus mykiss*) exhaustively exercised under normoxia (~100% air saturation) or hyperoxia (~200% air saturation).

All values are means $\pm$ S.E.M. (N=9 normoxia and N=10 hyperoxia). Cardiorespiratory responses are shown at standard metabolic rate (SMR; panels A-E) and maximum O₂ consumption rate (MO_{2max}; panels F-J). Scope = variable at MO_{2max} – variable at SMR (panels K-O). MO₂=mass-specific O₂ consumption rate; A-V O₂ content=arterial-venous O₂ content difference (estimated by the Fick equation). Asterisks show a significant difference (P<0.05) between normoxia and hyperoxia.

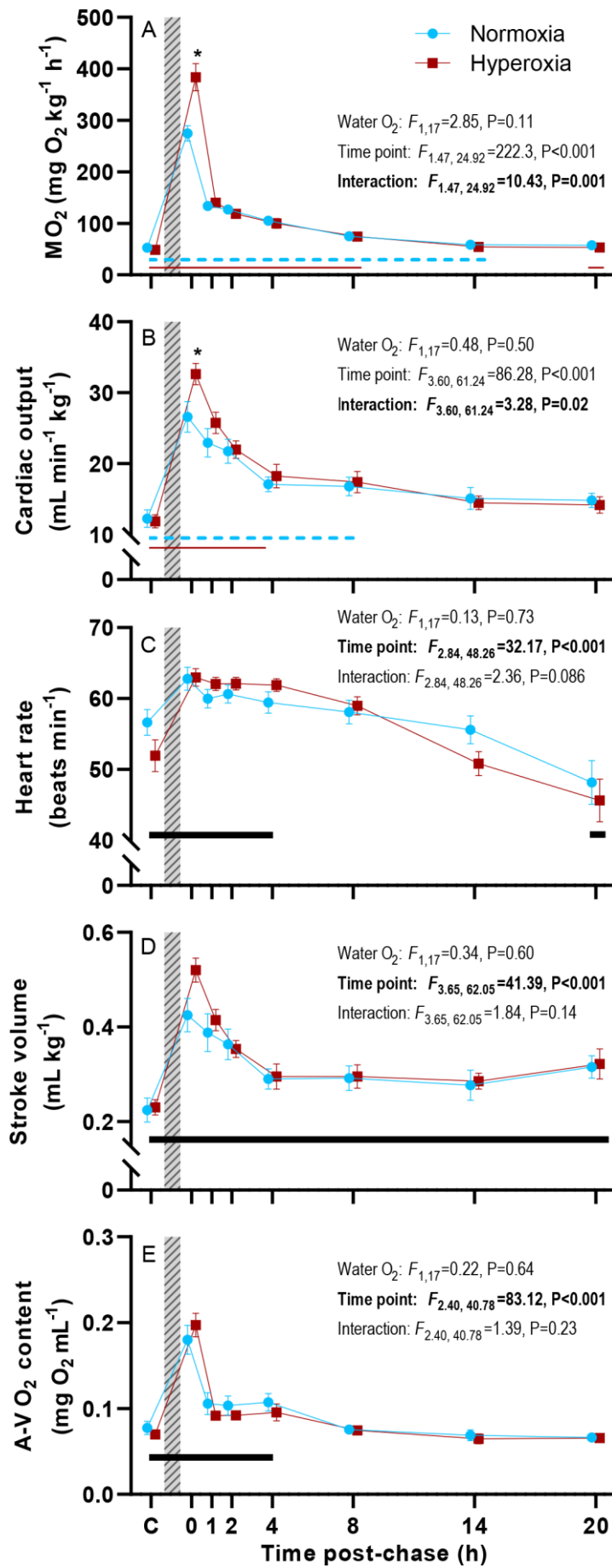


Fig. 2 Routine cardiorespiratory function in rainbow trout (*Oncorhynchus mykiss*) exhaustively exercised under normoxia (~100% air saturation) or hyperoxia (~200% air saturation). All values are means $\pm$ S.E.M. (N=9 normoxia and N=10 hyperoxia). A= $\dot{M}O_2$ (mass-specific O_2 consumption rate); B=Cardiac output; C=Heart rate; D=Cardiac stroke volume; E=Arterial-venous O_2 content difference (A-V O_2 content; estimated by the Fick equation). The hatched, grey bar represents 5 min exhaustive exercise (chasing) separating pre-chase control values (C on the x-axis) from post-chase values (0-20 h). Note, data points are intentionally staggered on the x-axis for ease of interpretation. Text insets show the results of a two-way mixed analysis of variance between water O_2 level and sample time point for each variable. Asterisks show significant differences ($P<0.05$) between normoxia and hyperoxia within a sampling time point. Dashed, blue and thin, maroon horizontal lines show significant differences ($P<0.05$) between pre-chase control values and post-chase values within the normoxia and hyperoxia treatments, respectively. Thick, black horizontal lines indicate where post-chase values were significantly different ($P<0.05$) from pre-chase control values across water O_2 levels (*i.e.*, where there was only a significant main effect of sampling time point).

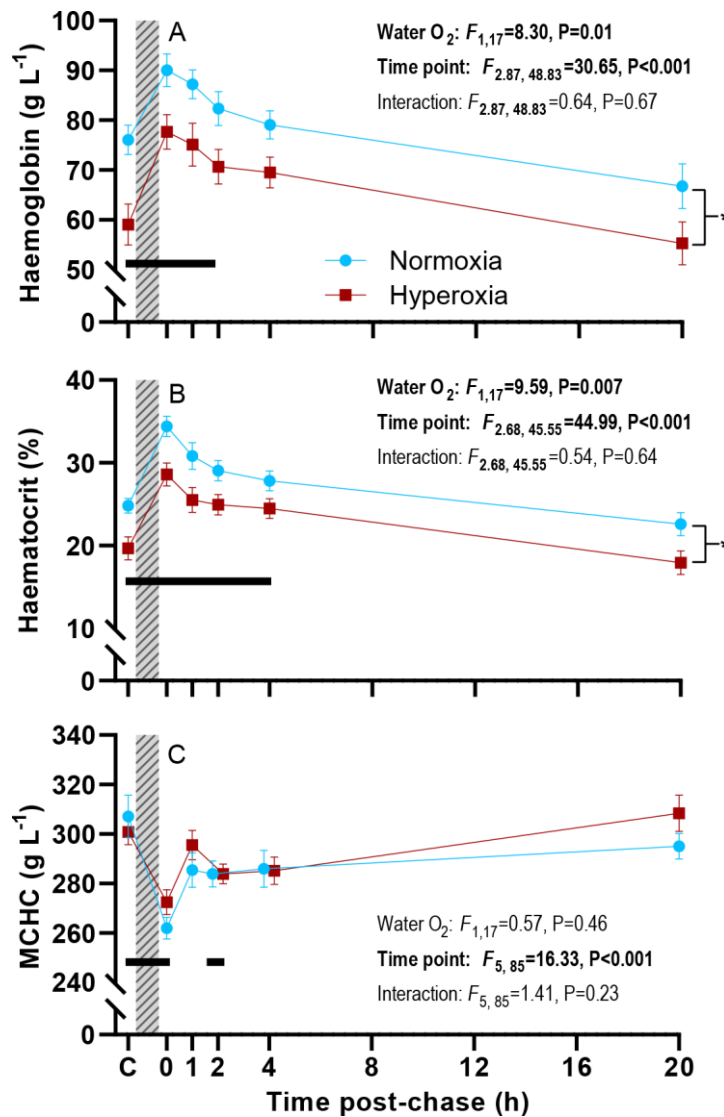


Fig. 3 Hematological responses in rainbow trout (*Oncorhynchus mykiss*) exhaustively exercised under normoxia (~100% air saturation) or hyperoxia (~200% air saturation). All values are means $\pm$ S.E.M. (N=9 normoxia and N=10 hyperoxia). A=Haemoglobin; B=Haematocrit; C=Mean corpuscular haemoglobin concentration (MCHC). Note, data points at 2 h and 4 h on panel C are intentionally staggered on the x-axis for ease of interpretation. The hatched, grey bar represents 5 min exhaustive exercise (chasing) separating pre-chase control values (C on the x-axis) from post-chase values (0-20 h). Text insets show the results of a two-way mixed analysis of variance between water O₂ level and sampling time point for each variable. Asterisks show a significant difference ($P<0.05$) between normoxia and hyperoxia across sampling time points (*i.e.*, a significant main effect of water O₂ level). Black horizontal lines indicate where post-chase values were significantly different ($P<0.05$) from pre-chase control values across water O₂ levels.

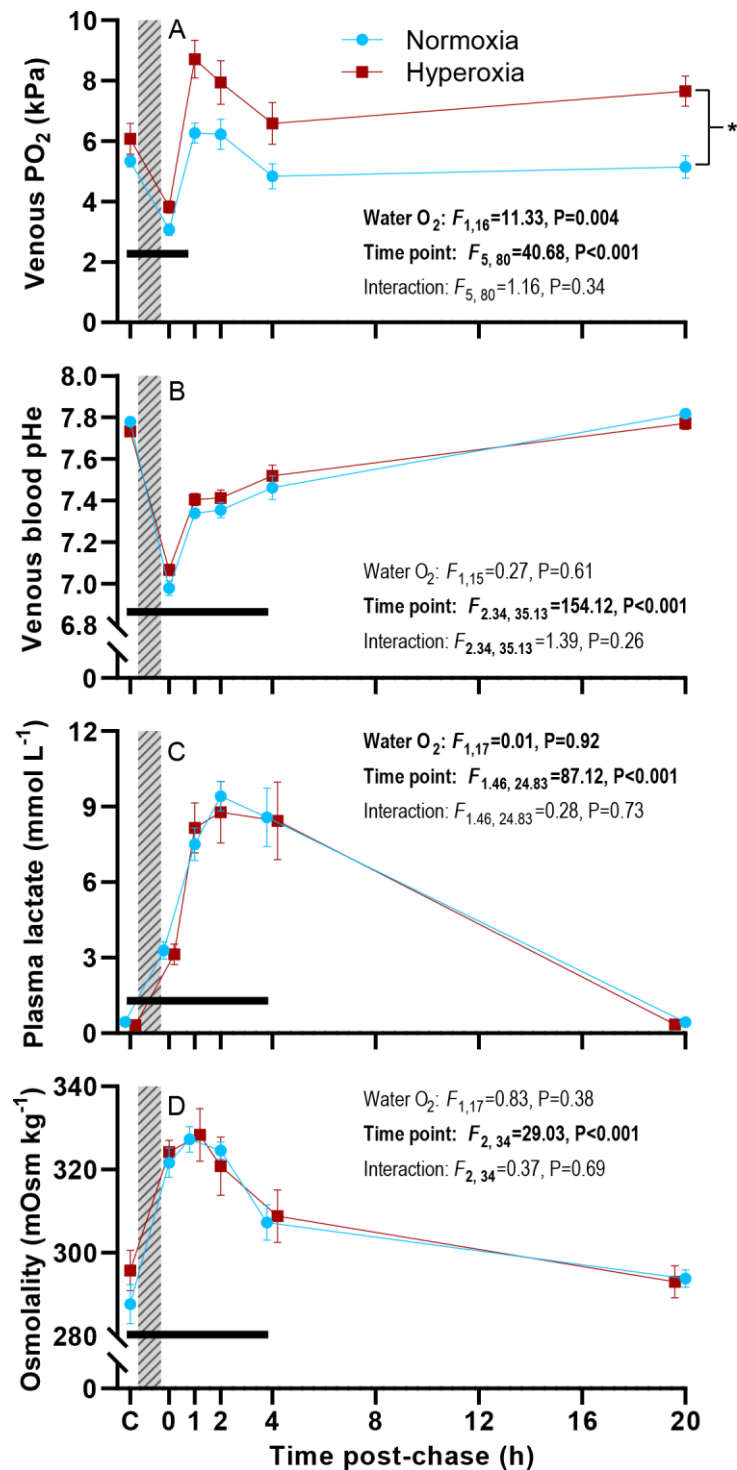


Fig. 4 Venous blood O₂ partial pressure, venous acid-base status, plasma lactate and plasma osmolality in rainbow trout (*Oncorhynchus mykiss*) exhaustively exercised under normoxia (~100% air saturation) or hyperoxia (~200% air saturation). All values are means $\pm$ S.E.M. (N=9 normoxia and N=10 hyperoxia). A=Venous blood O₂ partial pressure (PO₂); B=Venous blood extracellular pH (pHe); C=Plasma lactate; D=Plasma osmolality.

Note, some data points are intentionally staggered on the x-axis for ease of interpretation. The hatched grey bar represents 5 min exhaustive exercise (chasing) separating pre-chase control values (C on the x-axis) from post-chase values (0-20 h). Text insets show the results of a two-way mixed analysis of variance between water O₂ level and sample time point for each variable. Asterisks show a significant difference ($P < 0.05$) between normoxia and hyperoxia across sampling time points (*i.e.*, a significant main effect of water O₂ level). Black horizontal lines indicate where post-chase values were significantly different ($P < 0.05$) from pre-chase control values across water O₂ levels.

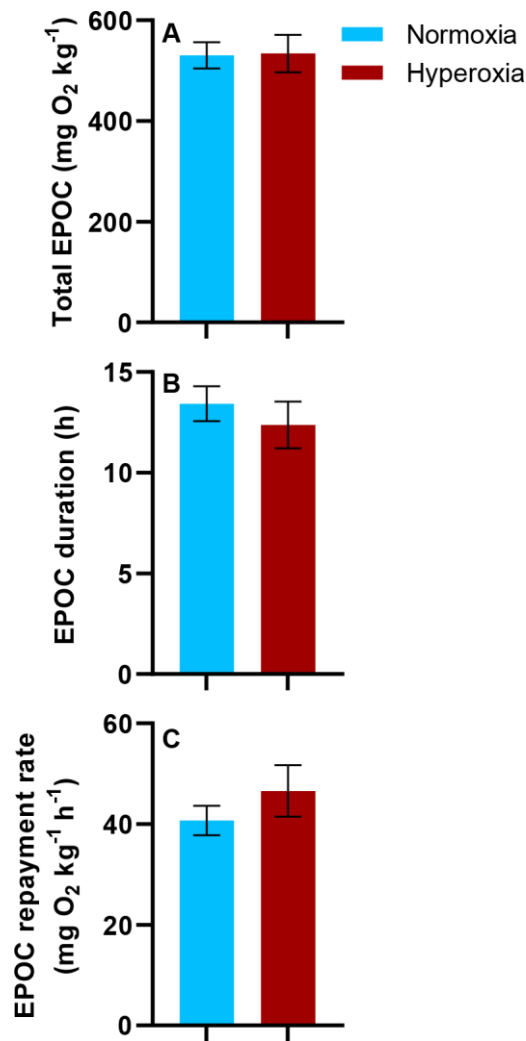


Fig. 5 Excess post-exercise oxygen consumption (EPOC) in rainbow trout (*Oncorhynchus mykiss*) exhaustively exercised under normoxia (~100% air saturation) or hyperoxia (~200% air saturation). All values are means $\pm$ S.E.M. (N=9 normoxia and N=10 hyperoxia). A=total EPOC, B=EPOC duration, and C=rate of EPOC repayment (*i.e.*, total EPOC/EPOC duration).